ECE 332: Engineering Electromagnetics II

Lecture 11

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May 5, 2026

Last Lecture

- The case of two transmission lines – a lossless line (Line 1) and an infinitely-long lossless line (Line 2) that meet at $z = 0$, is analogous to the case of a normally incident EM wave at a boundary between materials with different EM properties.
- The transmission-line quantities $(\tilde{V}, \tilde{I}, \beta, Z_0)$ and plane-wave quantities $(\tilde{E}, \tilde{H}, k, \eta)$ **satisfy the same equations**, respectively.

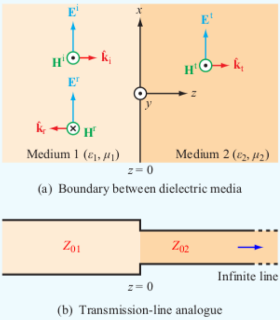


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Last Lecture (Continued)

- Lessons learned for transmission lines in Chapter 2 can be directly applied to normally-incident EM waves.

Table 8-1 Analogy between plane-wave equations for normal incidence and transmission-line equations, both under lossless conditions.

Plane Wave [Fig. 8-4(a)]		Transmission Line [Fig. 8-4(b)]	
$\widetilde{\mathbf{E}}_1(z) = \hat{\mathbf{x}} E_0^{\mathrm{i}} (e^{-jk_1 z} + \Gamma e^{jk_1 z})$	(8.11a)	$\widetilde{V}_1(z) = V_0^+ (e^{-j\beta_1 z} + \Gamma e^{j\beta_1 z})$	(8.11b)
$\widetilde{\mathbf{H}}_1(z) = \hat{\mathbf{y}} \dfrac{E_0^{\mathrm{i}}}{\eta_1} (e^{-jk_1 z} - \Gamma e^{jk_1 z})$	(8.12a)	$\widetilde{I}_1(z) = \dfrac{V_0^+}{Z_{01}} (e^{-j\beta_1 z} - \Gamma e^{j\beta_1 z})$	(8.12b)
$\widetilde{\mathbf{E}}_2(z) = \hat{\mathbf{x}} \tau E_0^{\mathrm{i}} e^{-jk_2 z}$	(8.13a)	$\widetilde{V}_2(z) = \tau V_0^+ e^{-j\beta_2 z}$	(8.13b)
$\widetilde{\mathbf{H}}_2(z) = \hat{\mathbf{y}} \tau \dfrac{E_0^{\mathrm{i}}}{\eta_2} e^{-jk_2 z}$	(8.14a)	$\widetilde{I}_2(z) = \tau \dfrac{V_0^+}{Z_{02}} e^{-j\beta_2 z}$	(8.14b)
$\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$		$\Gamma = (Z_{02} - Z_{01})/(Z_{02} + Z_{01})$	
$\tau = 1 + \Gamma$		$\tau = 1 + \Gamma$	
$k_1 = \omega\sqrt{\mu_1 \varepsilon_1}\,, \quad k_2 = \omega\sqrt{\mu_2 \varepsilon_2}$		$\beta_1 = \omega\sqrt{\mu_1 \varepsilon_1}\,, \quad \beta_2 = \omega\sqrt{\mu_2 \varepsilon_2}$	
$\eta_1 = \sqrt{\mu_1/\varepsilon_1}\,, \quad \eta_2 = \sqrt{\mu_2/\varepsilon_2}$		Z_{01} and Z_{02} depend on transmission-line parameters	

Last Lecture (Continued)

- If the characteristic impedances of Line 1 and Line 2 are Z_{01} and Z_{02}, resp., the **voltage reflection coefficient** at $z = 0$ from Line 1 to Line 2

$$\Gamma = \frac{Z_{02} - Z_{01}}{Z_{02} + Z_{01}} \tag{1}$$

  is analogous to the **reflection coefficient** Γ for the EM wave.

- The incident and reflected waves create a **standing wave pattern** in Medium 1, described by the **standing-wave ratio**

$$S = \frac{|\tilde{E}_1|_{\max}}{|\tilde{E}_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}. \tag{2}$$

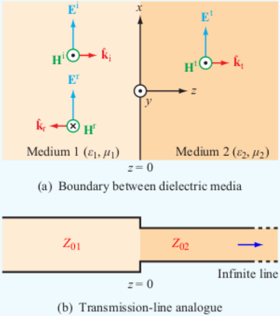


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Last Lecture (Continued)

- The **locations** $\ell_{\max}$ where the **electric field intensity reaches its maximum value** in Medium 1 are

$$\ell_{\max} = -z = \frac{\theta_\Gamma + 2n\pi}{2k_1} = \frac{\theta_\Gamma \lambda_1}{4\pi} + \frac{n\lambda_1}{2}, \tag{3}$$

where

$$\Gamma = |\Gamma| e^{j\theta_\Gamma} \text{ with } \theta_\Gamma \in (-\pi, \pi], \tag{4}$$

$$\lambda_1 = 2\pi/k_1, \tag{5}$$

and

$$n = \begin{cases} 1, 2, \cdots & \theta_\Gamma < 0 \\ 0, 1, 2, \cdots & \theta_\Gamma \geq 0. \end{cases} \tag{6}$$

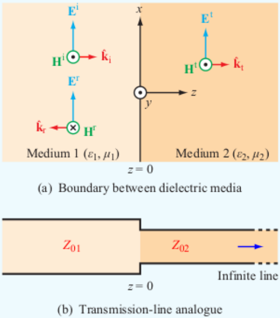


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Last Lecture (Continued)

- The **locations** ℓ_{min} where the **electric field intensity reaches its minimum value** in Medium 1 are

$$\ell_{\text{min}} = \begin{cases} \ell_{\text{max}} + \lambda_1/4, & \ell_{\text{max}} < \lambda_1/4 \\ \ell_{\text{max}} - \lambda_1/4, & \ell_{\text{max}} \geq \lambda_1/4. \end{cases} \tag{7}$$

- When the media are **lossless**,

$$\theta_\Gamma = \begin{cases} 0, & \eta_2 > \eta_1 \\ \pi, & \eta_2 < \eta_1. \end{cases} \tag{8}$$

- The expressions for ℓ_{max} and ℓ_{min} are also valid when Medium 1 is a **low-loss dielectric** and Medium 2 is a **dielectric or conductor**.

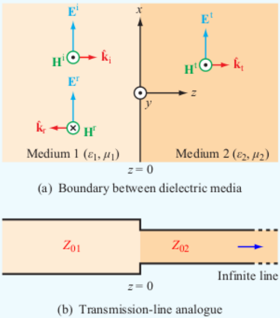


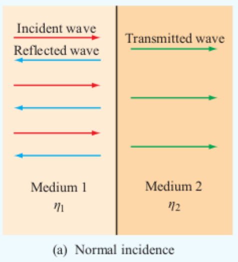
Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Power Flow in Lossless Media

We continued discussing what happens when an **x-polarized EM wave** traveling in the $+z$ direction **normal** to the **boundary between two lossless media** intersects the boundary, which is located at the x-y plane, and is **partially transmitted** and **partially reflected**.

Medium 1

- Fills half-space $z \leq 0$
- Permittivity ϵ_1
- Permeability μ_1
- $k_1 = \omega\sqrt{\mu_1 \epsilon_1}$, $\eta_1 = \sqrt{\mu_1/\epsilon_1}$



(a) Normal incidence

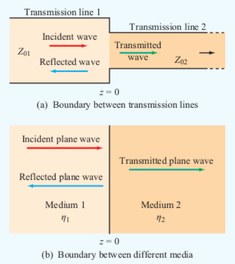


Figure 8-2 Discontinuity between two different transmission lines is analogous to that between two dissimilar media.

Last Lecture (Continued)

Medium 2

- Fills half-space $z \geq 0$
- Permittivity ϵ_2
- Permeability μ_2
- $k_2 = \omega\sqrt{\mu_2\epsilon_2}$, $\eta_2 = \sqrt{\mu_2/\epsilon_2}$

Incident Wave $(\mathbf{E}^i, \mathbf{H}^i)$

$$\tilde{\mathbf{E}}^i(z) = E_0^i e^{-jk_1 z}\hat{\mathbf{x}} \tag{9}$$

$$\tilde{\mathbf{H}}^i(z) = (E_0^i/\eta_1)e^{-jk_1 z}\hat{\mathbf{y}} \tag{10}$$

Reflected Wave $(\mathbf{E}^r, \mathbf{H}^r)$

$$\tilde{\mathbf{E}}^r(z) = \Gamma E_0^i e^{jk_1 z}\hat{\mathbf{x}} \tag{11}$$

$$\tilde{\mathbf{H}}^r(z) = -(\Gamma E_0^i/\eta_1)e^{jk_1 z}\hat{\mathbf{y}} \tag{12}$$

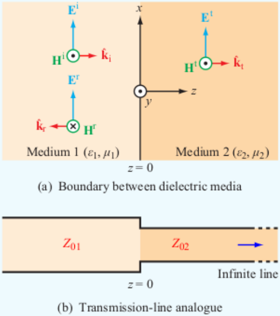


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Transmitted Wave $(\mathbf{E}^t, \mathbf{H}^t)$

$$\tilde{\mathbf{E}}^t(z) = \tau E_0^i e^{-jk_2 z}\hat{\mathbf{x}} \tag{13}$$

$$\tilde{\mathbf{H}}^t(z) = (\tau E_0^i/\eta_2)e^{-jk_2 z}\hat{\mathbf{y}} \tag{14}$$

Here, the **reflection coefficient** is

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \tag{15}$$

and the **transmission coefficient** is

$$\tau = \frac{2\eta_2}{\eta_2 + \eta_1} = 1 + \Gamma. \tag{16}$$

When the media are **lossless**, Γ and τ are **real-valued**.

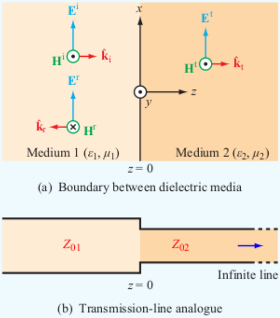


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Last Lecture (Continued)

The **electric** and **magnetic fields** in **Medium 1** are

$$\tilde{\mathbf{E}}_1(z) = E_0^i \left(e^{-jk_1 z} + \Gamma e^{jk_1 z} \right) \hat{\mathbf{x}} \qquad (17)$$

$$\tilde{\mathbf{H}}_1(z) = \frac{E_0^i}{\eta_1} \left(e^{-jk_1 z} - \Gamma e^{jk_1 z} \right) \hat{\mathbf{y}} \qquad (18)$$

and the **average power density** in **Medium 1** is

$$\begin{aligned} \mathbf{S}_{\text{av}1} &= \mathbf{S}_{\text{av}}^i + \mathbf{S}_{\text{av}}^r \\ &= \frac{\left|E_0^i\right|^2}{2\eta_1} \left(1 - |\Gamma|^2 \right) \hat{\mathbf{z}}. \qquad (19) \end{aligned}$$

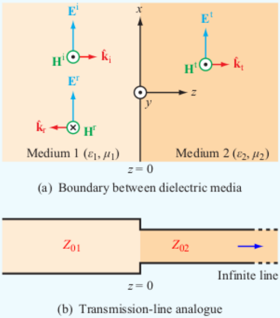


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

The **average power density** in **Medium 2** is

$$\mathbf{S}_{\text{av2}} = \frac{|\tau|^2 \left|E_0^i\right|^2}{2\eta_2}\hat{\mathbf{z}}. \tag{20}$$

Since (τ and Γ are real-valued and)

$$\frac{\tau^2}{\eta_2} = \frac{1-\Gamma^2}{\eta_1}, \tag{21}$$

$$\mathbf{S}_{\text{av2}} = \frac{\tau^2}{\eta_2}\frac{\left|E_0^i\right|^2}{2}\hat{\mathbf{z}} = \left(\frac{1-\Gamma^2}{\eta_1}\right)\frac{\left|E_0^i\right|^2}{2}\hat{\mathbf{z}} = \mathbf{S}_{\text{av1}}, \tag{22}$$

as expected from **power conservation**.

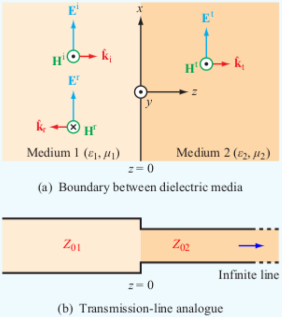


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Last Lecture (Continued)

- We explored an example of yellow light normally incident on glass.

Some possible applications of this case are

- windows
- lenses

Note that η_1, η_2, Γ, and τ are **independent of frequency** for **lossless media**. However, k_1 and k_2 are **frequency dependent** and $\mathbf{S}_{\text{av}1}$ and $\mathbf{S}_{\text{av}2}$ only depend on frequency through $|E_0^i|$.

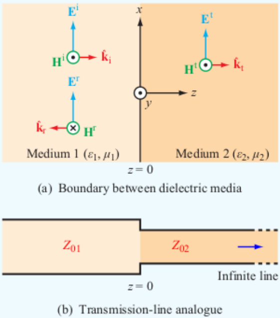


Figure 8-4 The two dielectric media separated by the x–y plane in (a) can be represented by the transmission-line analogue in (b).

Normal Incidence in Lossy Media

In a **lossy medium**, the **intrinsic impedance** η_c and **propagation constant** $\gamma = \alpha + j\beta$ are **complex-valued**.

Medium 1: ϵ_1, μ_1, σ_1,
$\eta_{c1} = \epsilon_1 - j\sigma_1/\omega$, $\gamma_1 = \alpha_1 + j\beta_1$

$$\tilde{\mathbf{E}}_1(z) = E_0^i \left(e^{-\gamma_1 z} + \Gamma e^{\gamma_1 z}\right) \hat{\mathbf{x}} \qquad (23)$$

$$\tilde{\mathbf{H}}_1(z) = \frac{E_0^i}{\eta_{c1}} \left(e^{-\gamma_1 z} - \Gamma e^{\gamma_1 z}\right) \hat{\mathbf{y}} \qquad (24)$$

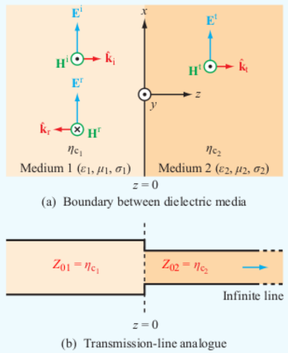


Figure 8-7 Normal incidence at a planar boundary between two lossy media.

Medium 2: ϵ_2, μ_2, σ_2,
$\eta_{c2} = \epsilon_2 - j\sigma_2/\omega$, $\gamma_2 = \alpha_2 + j\beta_2$

$$\tilde{\mathbf{E}}_2(z) = \tau E_0^i e^{-\gamma_2 z}\hat{\mathbf{x}} \tag{25}$$

$$\tilde{\mathbf{H}}_2(z) = \frac{\tau E_0^i}{\eta_{c2}} e^{-\gamma_2 z}\hat{\mathbf{y}} \tag{26}$$

As before, the **reflection and transmission coefficients** are

$$\Gamma = \frac{\eta_{c2} - \eta_{c1}}{\eta_{c2} + \eta_{c1}} \tag{27}$$

and

$$\tau = 1 + \Gamma = \frac{2\eta_{c2}}{\eta_{c2} + \eta_{c1}}. \tag{28}$$

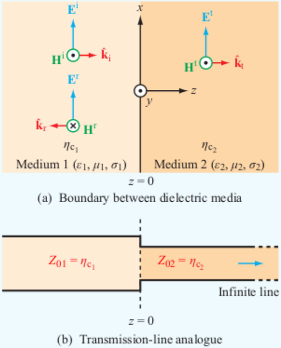


Figure 8-7 Normal incidence at a planar boundary between two lossy media.

Last Lecture (Continued)

- We considered an example of radio waves normally incident on a metal surface.

Some possible applications of this case are

- mirrors
- shielding
- filters

Note that η_{c1}, η_{c2}, Γ, τ, γ_1, and γ_2 are **frequency dependent** for **lossy media**.

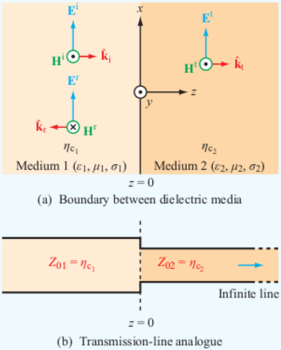


Figure 8-7 Normal incidence at a planar boundary between two lossy media.

Last Lecture (Continued)

Snell's Laws

For **lossless** media, when the incident wave is traveling in a direction that isn't normal to the boundary, the rays of the waves **travel in the directions below** and **intersect the boundary at the given angles relative to the normal to the boundary**.

- **incident wave:** direction $\hat{\mathbf{k}}_i$, angle $\theta_i \in [0, \pi/2]$,
- **reflected wave:** direction $\hat{\mathbf{k}}_r$, angle $\theta_r \in [0, \pi/2]$,
- **transmitted wave:** direction $\hat{\mathbf{k}}_t$, angle $\theta_t \in [0, \pi/2]$.

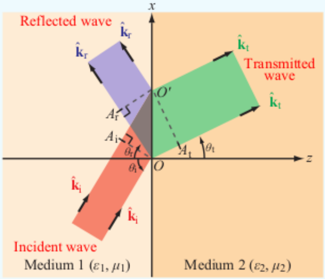
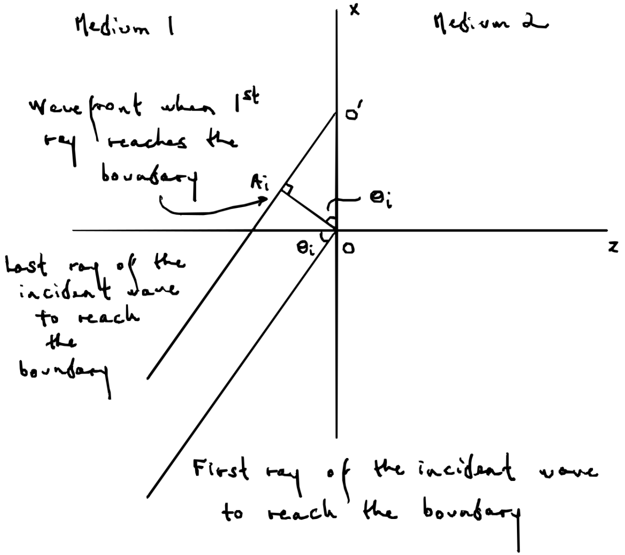
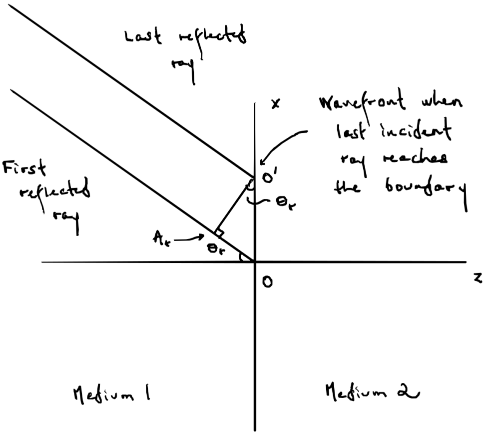


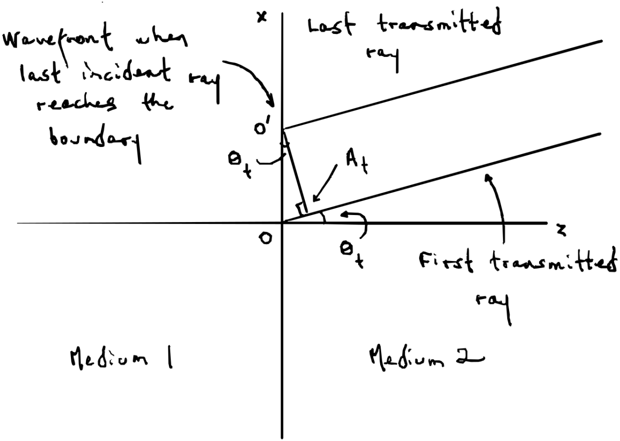
Figure 8-9 Wave reflection and refraction at a planar boundary between different media.

Last Lecture (Continued)





The time it takes for the last ray of the incident wave to reach the boundary to go from point A_i to point O' is the same as the time the first reflected ray travels from point O to point A_r.



The time it takes for the last ray of the incident wave to reach the boundary to go from point A_i to point O' is the same as the time the first transmitted ray travels from point O to point A_t.

Last Lecture (Continued)

- Under these circumstances,

$$\theta_r = \theta_i \quad \text{(Snell's law of } \textbf{reflection}\text{)} \tag{29}$$

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1\epsilon_1}{\mu_2\epsilon_2}} = \sqrt{\frac{\mu_{r_1}\epsilon_{r_1}}{\mu_{r_2}\epsilon_{r_2}}} \quad \text{(Snell's law of } \textbf{refraction}\text{).} \tag{30}$$

Snell's Laws (Continued)

The **index of refraction** of a medium with phase velocity u_p is

$$n = \frac{c}{u_p} = \sqrt{\frac{\mu\epsilon}{\mu_0\epsilon_0}} = \sqrt{\mu_r\epsilon_r}. \tag{31}$$

Therefore, altogether

$$\theta_r = \theta_i \quad \text{(Snell's law of \textbf{reflection})} \tag{32}$$

$$\frac{\sin\theta_t}{\sin\theta_i} = \frac{n_1}{n_2} = \frac{u_{\text{p}2}}{u_{\text{p}1}} = \sqrt{\frac{\mu_1\epsilon_1}{\mu_2\epsilon_2}} = \sqrt{\frac{\mu_{r_1}\epsilon_{r_1}}{\mu_{r_2}\epsilon_{r_2}}} \quad \text{(Snell's law of \textbf{refraction})}. \tag{33}$$

For **nonmagnetic materials**, $\mu_1 = \mu_2 = \mu_0$ and the **law of refraction** is

$$\frac{\sin\theta_t}{\sin\theta_i} = \sqrt{\frac{\epsilon_{r_1}}{\epsilon_{r_2}}} = \frac{\eta_2}{\eta_1}. \tag{34}$$

Denser nonmagnetic materials often have higher permittivities and, hence, higher indices of refraction. So, nonmagnetic materials with **higher indices of refraction** are often called **denser** compared to nonmagnetic materials with **lower indices of refraction**.

As a sanity check, suppose $\theta_i = 0$. This is the case of **normal incidence**. In this case,

$$\theta_r = \theta_i = 0 \tag{35}$$

and

$$\sin\theta_t = \frac{n_1}{n_2}\sin\theta_i = 0 \implies \theta_t = 0, \tag{36}$$

regardless of the relationship between n_1 and n_2, as expected.

More generally, when $0 < \theta_i \leq \pi/2$ and $n_2 > n_1 > 0$,

$$\sin\theta_t = \frac{n_1}{n_2}\sin\theta_i < \sin\theta_i \implies \theta_t < \theta_i \tag{37}$$

whereas, when $0 < \theta_i \leq \pi/2$ and $n_1 > n_2 > 0$,

$$\sin\theta_t = \frac{n_1}{n_2}\sin\theta_i > \sin\theta_i \implies \theta_t > \theta_i \tag{38}$$

So, when $\theta_i \neq 0$ and $n_1 < n_2$, the transmitted wave is directed toward the normal to the boundary, which is called **inward refraction**, and, when $\theta_i \neq 0$ and $n_1 > n_2$, the transmitted wave is directed away from the normal to the boundary, which is called **outward refraction**.

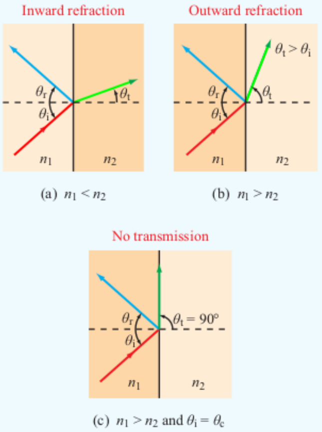


Figure 8-10 Snell's laws state that $\theta_r = \theta_i$ and $\sin\theta_t = (n_1/n_2)\sin\theta_i$. Refraction is (a) inward if $n_1 < n_2$ and (b) outward if $n_1 > n_2$; and (c) the refraction angle is 90° if $n_1 > n_2$ and θ_i is equal to or greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

When $n_1 > n_2$, there will be an angle θ_i where $\theta_t = \pi/2$. This is called the **critical angle** θ_c and it satisfies

$$\sin\theta_c = \frac{n_2}{n_1}\sin\left(\frac{\pi}{2}\right) = \frac{n_2}{n_1}. \qquad (39)$$

If $\mu_1 = \mu_2$,

$$\sin\theta_c = \sqrt{\frac{\epsilon_{r_2}}{\epsilon_{r_1}}}. \qquad (40)$$

When $n_1 > n_2$ and $\theta_i \geq \theta_c$, the incident wave is entirely reflected, which is called **total internal reflection**.

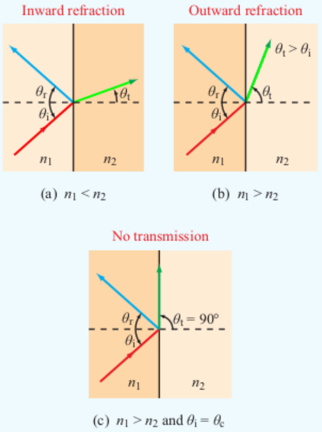


Figure 8-10 Snell's laws state that $\theta_r = \theta_i$ and $\sin\theta_t = (n_1/n_2)\sin\theta_i$. Refraction is (a) inward if $n_1 < n_2$ and (b) outward if $n_1 > n_2$; and (c) the refraction angle is 90° if $n_1 > n_2$ and θ_i is equal to or greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

Fiber Optics

Small-diameter cylinders of dielectric material, typically glass or plastic, can efficiently carry light via the phenomenon of total internal reflection.

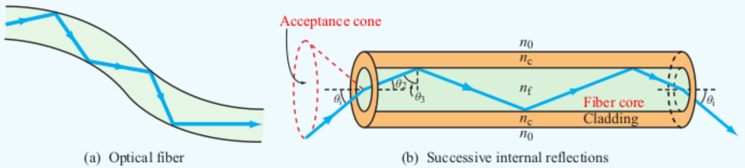


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Here, the **fiber core** has an index of refraction n_f that is higher than the index of refraction n_c of the surrounding **cladding**. The critical angle for the fiber is given by

$$\sin\theta_c = \frac{n_c}{n_f}. \tag{41}$$

A light ray traveling through the fiber will intersect the outer surface of the cylinder at an angle $\theta_3 \in (0, \pi/2]$ relative to the normal to the surface.

To ensure total internal reflection, θ_3 must satisfy

$$\sin\theta_3 \geq \sin\theta_c = \frac{n_c}{n_f}. \tag{42}$$

The angle that the ray makes with the central axis of the fiber is

$$\theta_2 = \frac{\pi}{2} - \theta_3. \tag{43}$$

So, for the light to be totally internally reflected, θ_2 must satisfy

$$\cos\theta_2 = \cos\left(\frac{\pi}{2} - \theta_3\right) = \sin\theta_3 \geq \frac{n_c}{n_f}. \tag{44}$$

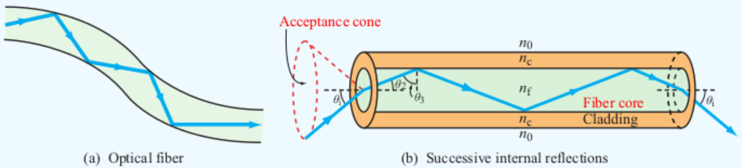


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

If the index of refraction of the medium outside of the fiber is n_0 and a ray of light shining onto the end of the fiber intersects the fiber core at an angle θ_i relative to the normal to the end (i.e. the direction of the central axis of the core), then, according to the law of refraction,

$$\sin\theta_2 = \frac{n_0}{n_f}\sin\theta_i. \tag{45}$$

By a trigonometric identity,

$$1 = \cos^2\theta_2 + \sin^2\theta_2 = \cos^2\theta_2 + \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i. \tag{46}$$

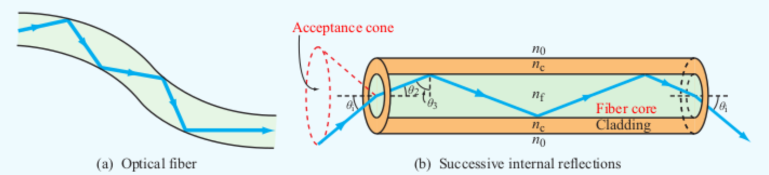


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Fiber Optics (Continued)

Solving for $\cos\theta_2$,

$$\cos\theta_2 = \left(1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i\right)^{1/2}. \tag{47}$$

So, for total internal reflection, θ_i must satisfy

$$\left(1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i\right)^{1/2} \geq \frac{n_c}{n_f}. \tag{48}$$

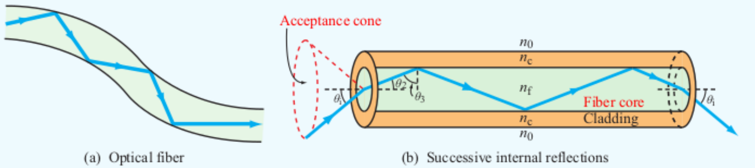


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Squaring both sides of this inequality,

$$1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i \geq \frac{n_c^2}{n_f^2} \implies \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i \leq 1 - \frac{n_c^2}{n_f^2} \tag{49}$$

so that

$$\sin^2\theta_i \leq \left(\frac{n_f}{n_0}\right)^2 \left(1 - \frac{n_c^2}{n_f^2}\right) = \frac{1}{n_0^2}\left(n_f^2 - n_c^2\right) \tag{50}$$

and, finally, taking the square root of both sides

$$\sin\theta_i \leq \frac{1}{n_0}\left(n_f^2 - n_c^2\right)^{1/2}. \tag{51}$$

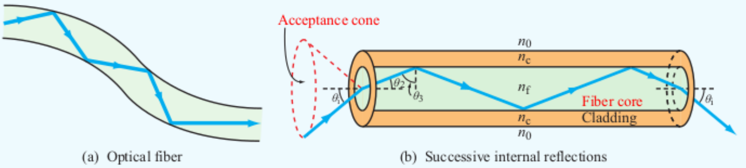


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Maximum Acceptance Cone

The **largest angle** θ_i that produces total internal reflection in the fiber is the **acceptance angle** θ_a, which satisfies

$$\sin\theta_a = \frac{1}{n_0}\left(n_f^2 - n_c^2\right)^{1/2}. \tag{52}$$

The cone containing all rays that intersect the fiber core with $\theta_i \leq \theta_a$ is called the **acceptance cone**.

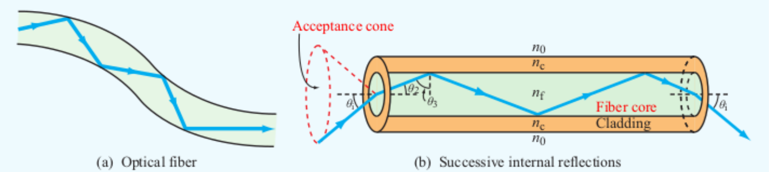


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Fiber Optics (Continued)

Rays entering the fiber core at different angles θ_i follow different paths through the fiber. Each path is called a **mode** and different modes travel through the fiber in different amounts of time. This is called **modal dispersion** and distorts pulses sent through the fiber. This causes rectangular pulses of initial width τ_i and period T to be spread out into rounded pulses of width τ. If $\tau > T$, the pulses arriving at the end of the fiber will overlap and it might not be possible to interpret the signal. To prevent this problem, the pulses are usually created with a period $T \geq 2\tau$.

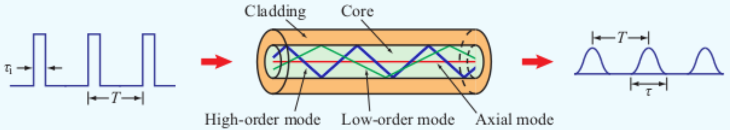


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

Fiber Optics (Continued)

Here,

$$\cos\theta_2 = n_c/n_f \tag{53}$$

and the maximum distance a ray will travel through a fiber of length ℓ is

$$\ell_{\max} = \frac{\ell}{\cos\theta_2} = \ell n_f/n_c \tag{54}$$

So, the travel time for the longest path will be

$$t_{\max} = \ell_{\max}/u_p = \frac{\ell n_f^2}{c n_c}. \tag{55}$$

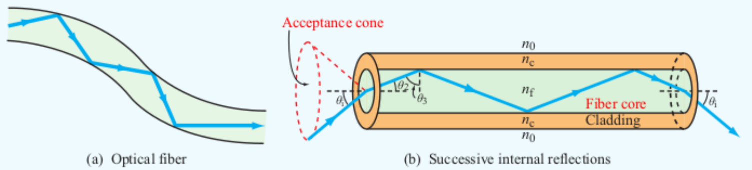


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

Fiber Optics (Continued)

The minimum travel time will be

$$t_{\text{min}} = \frac{\ell}{u_p} = \frac{\ell n_f}{c} \tag{56}$$

and the maximum time delay will be

$$\tau = \Delta t = t_{\text{max}} - t_{\text{min}} = \frac{\ell n_f}{c}\left(\frac{n_f - 1}{n_c}\right). \tag{57}$$

Requiring $T \geq 2\tau$, the **maximum data transmission rate** will then be

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{c n_c}{2\ell n_f (n_f - n_c)}. \tag{58}$$

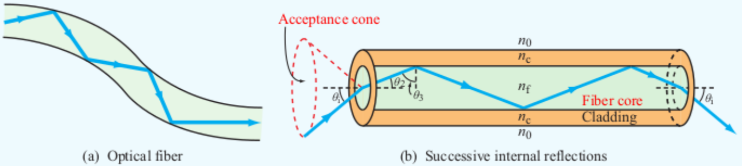


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

The cone consisting of all of the paths that light at the end of the fiber is allowed to follow is called the **entrance cone**.

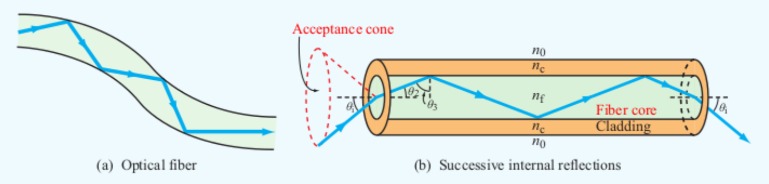


Figure 8-12 Waves can be guided along optical fibers as long as the reflection angles exceed the critical angle for total internal reflection.

If the entrance cone can be restricted to be narrower than the acceptance cone, with maximum angle $\theta_i(\text{max})$, then the **maximum data transmission rate** can be **increased** to

$$f_p = \frac{c}{2ln_f}\left(\frac{\cos\theta_2(\text{max})}{1-\cos\theta_2(\text{max})}\right), \tag{59}$$

Fiber Optics (Continued)

where

$$\cos\theta_2(\max) = \left(1 - \left(\frac{n_0}{n_f}\right)^2 \sin^2\theta_i(\max)\right)^{1/2}. \tag{60}$$

Homework

Read Chapter 8.

Exam 1 is on Thursday.

Homework 2 was due today.